



YOUR TECHNOLOGY & INNOVATION PARTNER

Advanced AI Training - Session 5

Domain-Specific Applications - Healthcare AI



Course Objectives

- Gain an overview of AI's evolution in healthcare.
- Explore fundamental AI techniques and models.
- Understand the processes for developing and fine-tuning healthcare AI solutions.
- Identify diverse applications and real-world projects of AI in healthcare.
- Appreciate the significant benefits of AI for various healthcare stakeholders.
- Recognize the challenges and limitations of AI implementation in healthcare.
- Learn about responsible AI considerations.
- Discover emerging trends and the future direction of AI in healthcare.
- Develop essential skills for building impactful AI healthcare solutions.

Table of Contents

Overview of AI in Healthcare

Healthcare AI Technologies

Healthcare AI Applications

Benefits of AI in Healthcare

Challenges and Limitations

Future of AI in Healthcare

Conclusion

Q&A

Overview of AI in Healthcare

Healthcare AI: Key Milestones

1956

AI Coined at Dartmouth

Term "artificial intelligence" coined, recognizing AI as a scientific field, laying foundational concepts.

Early 1970s

MYCIN developed

First expert system (MYCIN) for bacterial infection diagnosis.

1990s

Emergence of Machine Learning

Shift to data-driven machine learning, crucial for modern AI.

Late 2010s

AI integration with clinical practice

AI integrated into clinical practice with user-friendly CDSS and EHRs.

2012

AlexNet breakthrough

AlexNet breakthrough sparked deep learning advancements in medical imaging.

Early 2000s

Deep Learning advancements begin

Deep Learning advancements begin, surging AI utility in healthcare.

2020s-Present

Widespread AI adoption & GenAI

AI, especially generative AI, becomes integral to healthcare operations.

2023

FDA approval of AI

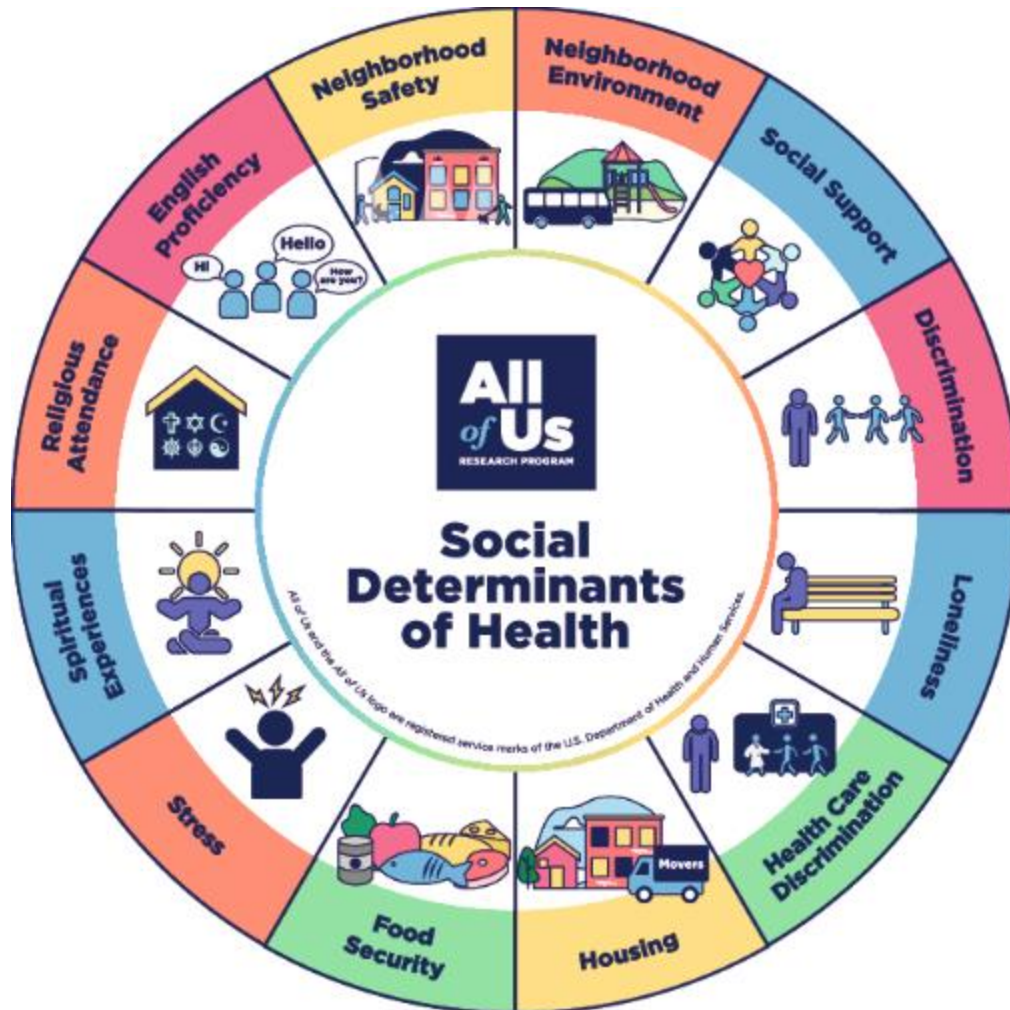
FDA approves 223 AI medical devices, showing maturation and validation.

2025

GenAI Transforms Clinical Workflows

Generative AI widely implemented in clinics, boosting efficiency and patient interaction.

The Data Foundation



Critical Data Types

- Clinical data (EHRs, images, labs)
- Claims and billing data
- Patient-generated health data (PGHD)
- Social determinants of health (SDOH)
- Data from Wearable Devices and IoT Sensors
- Research and trial data

Healthcare AI Technologies

AI Techniques



Machine Learning (ML)



Rule-Based Expert Systems



**Deep Learning &
Neural Networks**



Natural Language Processing (NLP)



Robotics & Automation



Generative AI

Healthcare AI Technologies



Foundation Models

LLMs fine-tuned for healthcare, multimodal AI, domain-specific models



Data Processing

NLP for clinical notes, computer vision for imaging, predictive analytics, knowledge graphs

Healthcare AI Models (1/3)



Meditron is a suite of open-source medical Large Language Models (LLMs)

source: <https://github.com/epfLLM/meditron>

▼ Advisory Notice

While Meditron is designed to encode medical knowledge from sources of high-quality evidence, it is not yet adapted to deliver this knowledge appropriately, safely, or within professional actionable constraints. We recommend against using Meditron in medical applications without extensive use-case alignment, as well as additional testing, specifically including randomized controlled trials in real-world practice settings.

Model Details

- Developed by: [EPFL LLM Team](#)
- Model type: Causal decoder-only transformer language model
- Language(s): English (mainly)
- Model License: [LLAMA 2 COMMUNITY LICENSE AGREEMENT](#)
- Code License: [APACHE 2.0 LICENSE](#)
- Continue-pretrained from model: [Llama-2-70B](#)
- Context length: 4k tokens
- Input: Text only data
- Output: Model generates text only
- Status: This is a static model trained on an offline dataset. Future versions of the tuned models will be released as we enhance model's performance.
- Knowledge Cutoff: August 2023
- Trainer: [epfLLM/Megatron-LLM](#)
- Paper: [Meditron-70B: Scaling Medical Pretraining for Large Language Models](#)

Healthcare AI Models (2/3)

The screenshot displays the Azure AI Studio Model catalog interface. The top navigation bar includes the Azure AI Studio logo, the 'Model catalog' tab, and user information for 'Contoso Responsible R&D' with a user ID of 'ContosoPharmaceutical-0034'. A left sidebar contains navigation links for 'Home', 'AI services' (with 'Model catalog' selected), 'Model benchmarking', 'Prompt Catalog', 'AI Services', and 'Management' (with 'All Hubs', 'Resources & keys', and 'Quota'). The main content area is titled 'Find the right model to build your custom AI solution' and features an 'Announcements' section. Below this is a filter bar with options for 'All filters', 'Publishers', 'Deployment options', 'Tasks', 'Licenses', and 'Industry (1)'. A search bar is also present. The models are displayed in a grid, each with a Microsoft or OpenAI logo, a name, and a description. The right sidebar contains a 'Filter by' section with expandable categories: 'Collections', 'Industry' (with 'Health & lifesciences' selected), 'Deployment options' (with 'Managed compute' and 'Serverless API' selected), 'Inference tasks', 'Fine-tuning tasks', and 'Licenses'. At the bottom of the grid, there are 'Prev' and 'Next' navigation arrows.

Find the right model to build your custom AI solution

Announcements

All filters Publishers Deployment options Tasks Licenses Industry (1)

Search

Models 16

microsoft-MedImageInsight Embeddings	microsoft-CXRReportGen Findings generation	microsoft-MedImageParse Segmentation
microsoft-gigapath Image feature extraction	m42-health-Llama3-Med42-70B Text generation	microsoft-biomednlp-pubmed... Fill mask
UFNLP-gatortron-base Text generation	microsoft-biomednlp-pubmed... Fill mask	microsoft-biomednlp-pubmed... Fill mask
microsoft-biogpt Text generation	microsoft-biogpt-large-pubme... Text generation	microsoft-biogpt-large Text generation
microsoft-biomedCLIP-PubMedBERT_2... Text generation	medical-ai-clinicalBERT Text generation	stanford-crfm-BioMedLM Text generation
BioMistral-7B Text generation		

Prev Next

Healthcare AI Models (3/3)

Search models
Q healthcare



Search results for "healthcare"

Sort by: **Trending** [Newest](#) [Last updated](#)

Name	Description	Type	Task	Modality	Resource ID	
Gemma	Lightweight, state-of-the-art open models from Google, built from the same research and technology used to create the...	Foundation	Generation	Language	gemma-2b	▼
PaliGemma 1 & 2	Lightweight, state-of-the-art open models from Google, built from the same research and technology used to create the...	Foundation	Generation	Multimodal	google/paligemma2-3b-...	▼
CodeGemma	Lightweight, state-of-the-art open models from Google, built from the same research and technology used to create the...	Foundation	Generation	Language	google/codegemma-7b-it	▼
Gemma 2	Lightweight, state-of-the-art open models from Google, built from the same research and technology used to create the...	Foundation	Generation	Language	gemma2-2b-it	▼
MedGemma	Collection of Gemma 3 variants that are trained for performance on medical text and image comprehension.	Foundation	Generation	Language	google/medgemma-4b-it	
Gemma 3	Lightweight, state-of-the-art open models from Google, built from the same research and technology used to create the...	Foundation	Generation	Language	gemma-3-1b-it	▼
ShieldGemma 2	Lightweight, state-of-the-art open models from Google, built from the same research and technology used to create the...	Foundation	Generation	Multimodal	shieldgemma-2-4b-it	▼
CXR Foundation	CXR Foundation produces embeddings based on images of chest X-rays that can be used to efficiently train classifier...	Foundation	Embedding	Vision	google/cxr-foundation	▼
Path Foundation	Path Foundation produces embeddings that can be used to efficiently train classifier models for pathology analysis tasks ...	Foundation	Embedding	Vision	google/path-foundation	▼
Derm Foundation	Derm Foundation produces embeddings that can be used to efficiently train classifier models for dermatology-related task...	Foundation	Embedding	Vision	google/derm-foundation	▼
TxGemma	TxGemma generates predictions, classifications or text based on therapeutic related data and can be used to efficiently...	Foundation	Generation	Language	google/txgemma-27b-...	▼
HeAR	HeAR produces embeddings based on audio clips of health-related sounds that can be used to efficiently train classifier...	Foundation	Embedding	Audio	google/hear	▼

Image Embedding

Using an image embedding model to search for similar images or facilitate detection of anomalies that could indicate potential data issues or system errors

Out-of-box embedding

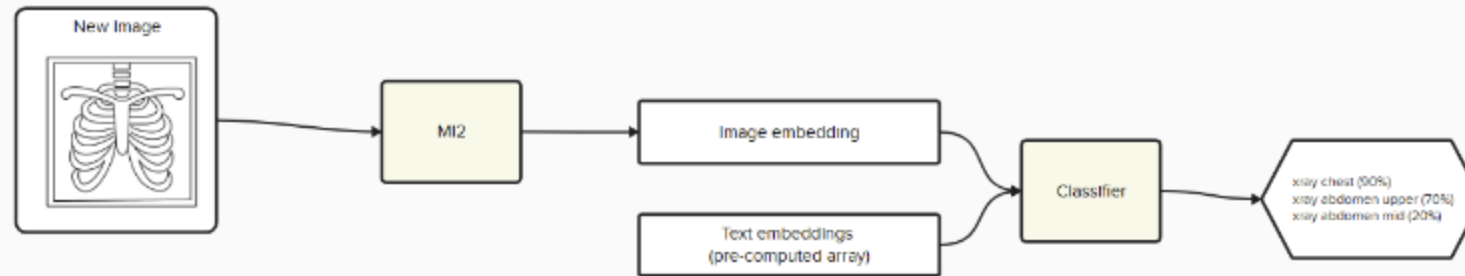
Adapter to Specific Task

Building an adapter to the embedding model for a specific task.

Adapter to specific task

Zero-Shot and Adapter Approaches for Medical Image

Zero-Shot Approach



Text-Image Correlations

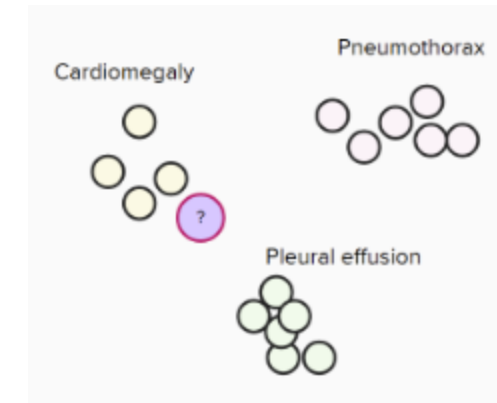
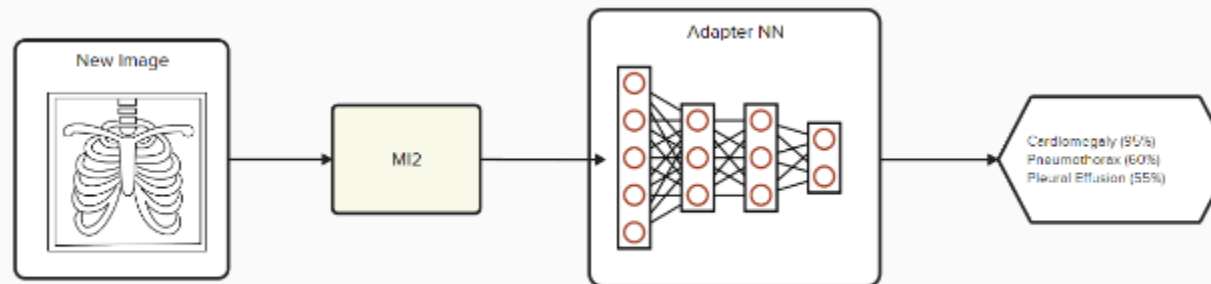


Image-Image Clustering

Adapter Approach

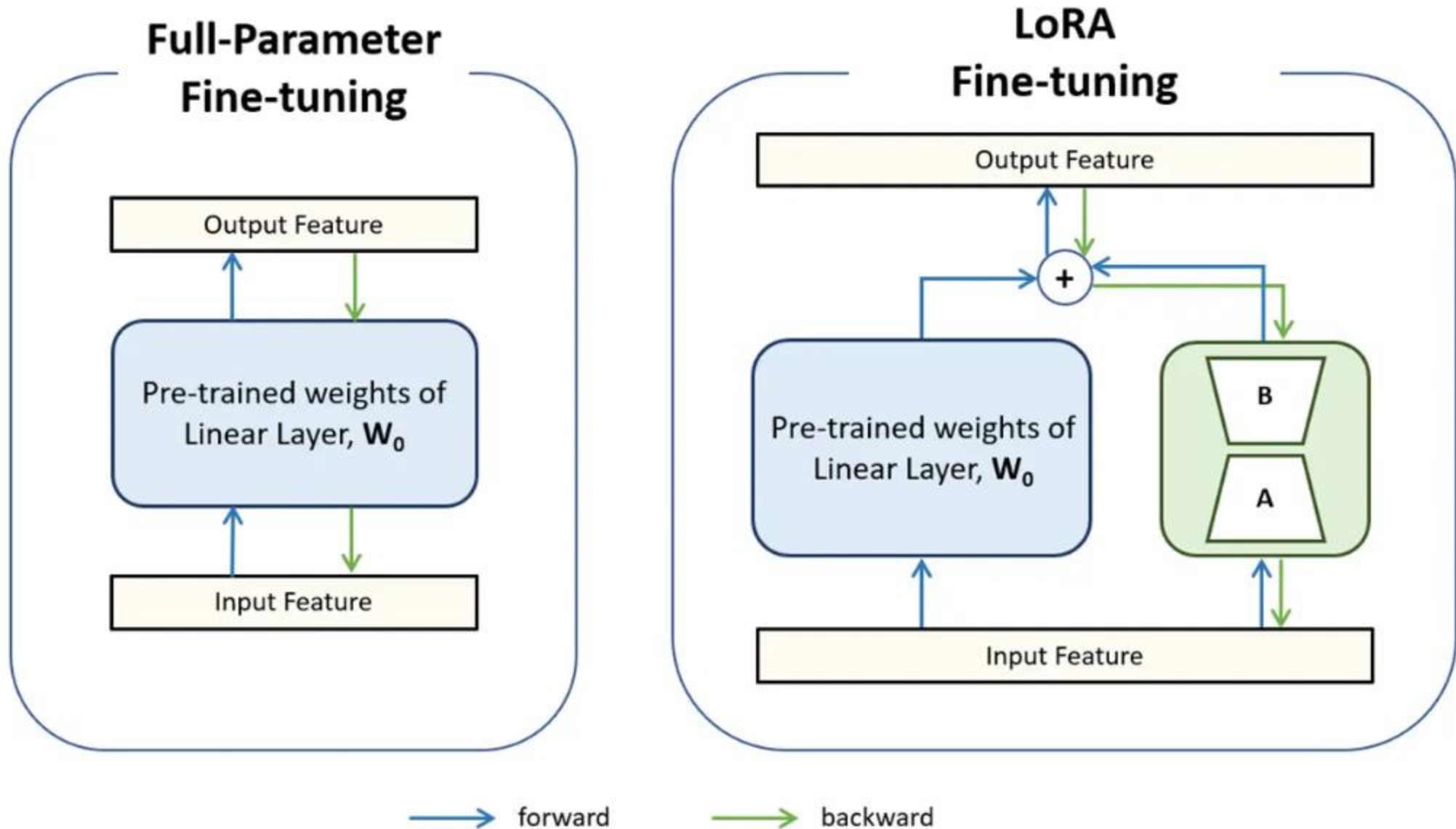


Fine-tuning for Specific Task

Fine-tuning pretrained unimodal models to create a narrow model.

Fine-tuning for specific task

Full-Parameter vs Lora Fine-tuning



Adapter to General Reasoner

Integrating language models to enable the extraction of insights across modalities and enhance the interpretability of multimodal data.

Adapter to general reasoner
(language model)

Connect Modalities

Connecting different data modalities for a more comprehensive, holistic view of data that derives new insights and enables the discovery of previously hidden correlations and patterns.

Connecting modalities



Healthcare AI Applications

Sample Solutions / Projects

AI Agent/Chatbot

1

Knowledge
Based Chatbot

2

Healthcare
Virtual Assistant

Behavior Analytics

3

Sleep Quality Analysis

Medical Image Analytics

4

Tooth Recognition &
Scoring

5

Lung Cancer Detection

Early Warning

6

Stroke Assessment and Detection

1. Knowledge Based Chatbot - TMA

Customer Needs:

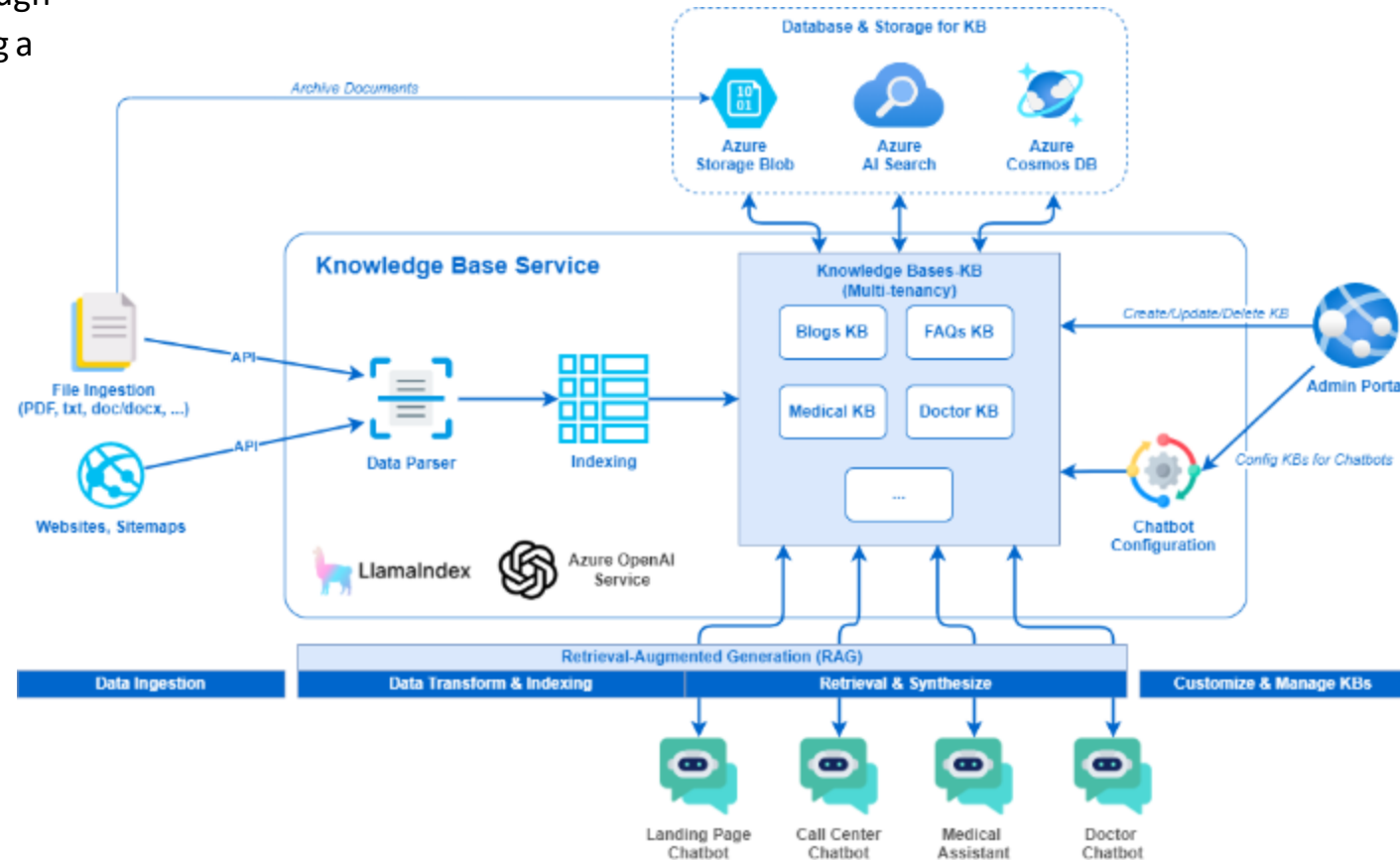
The client is a telehealth provider offering consultations through video, phone, and in-person visits, faced challenges managing a high volume of customers, long wait times on live chat and phone calls resulted in lost customers

Key Features & Benefits:

- AI Chatbot is developed to provide quick responses, automate appointment bookings, and enhancing user engagement
- Assists both patients and doctors, support in treatment programs and helps in training new doctors in the specific field

Technologies:

- KB Development: Document/Website/Sitemap Parser; RAG; Data Agents, LlamaIndex
- Azure AI Search, Azure OpenAI Service, Azure Cosmos DB for MongoDB, Azure Blob Storage
- FastAPI, Python



2. Healthcare Virtual Assistant - TMA

Project Description:

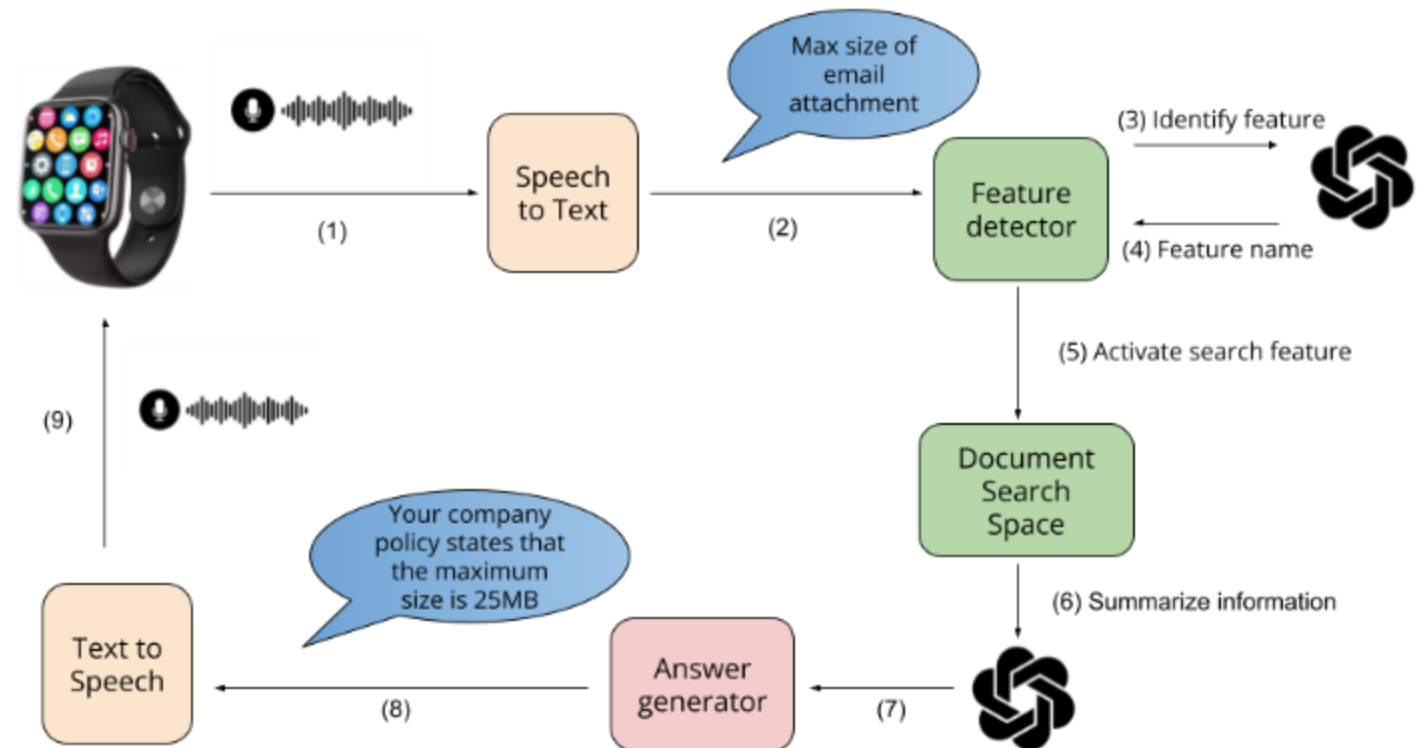
This project aims to leverage voice interaction and GPT model on wearable devices to identify user intent and activate the corresponding product features.

TMA Values and Achievements:

- Assistant can provide answers for both organization-related topics, such as rules, policies, and general topics.
- Be able to deploy on many smart watches

Technologies:

- OpenAI, ChatGPT
- Speech2Text, Text2Speech SDK
- Natural language processing



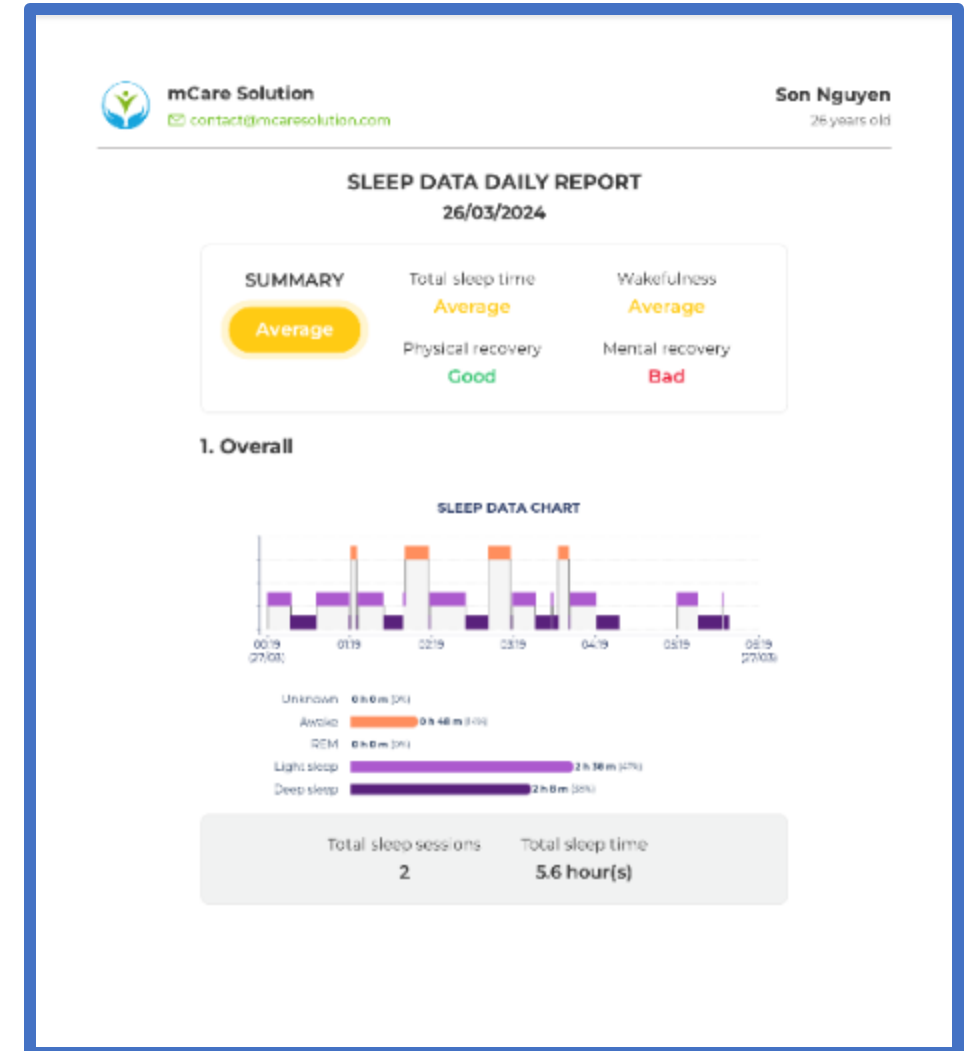
3. Sleep Quality Analysis - TMA

Enhances sleep health using machine learning to analyze wearable device data, patient reports, and sleep studies, detecting disorders like sleep apnea and insomnia, offering personalized sleep plans, and tracking treatment effectiveness.

Data: Wearable device sleep data (heart rate, SPO2, REM, light/deep sleep), patient-reported data.

Method: Machine learning analyzes sleep stage patterns, AI predicts trends using heart rate and SPO2 during sleep.

Applications: Detects sleep disorders, provides personalized sleep improvement plans, monitors treatment effectiveness, and improves mental and physical health.



4. Tooth Recognition & Scoring - TMA

Customer Needs:

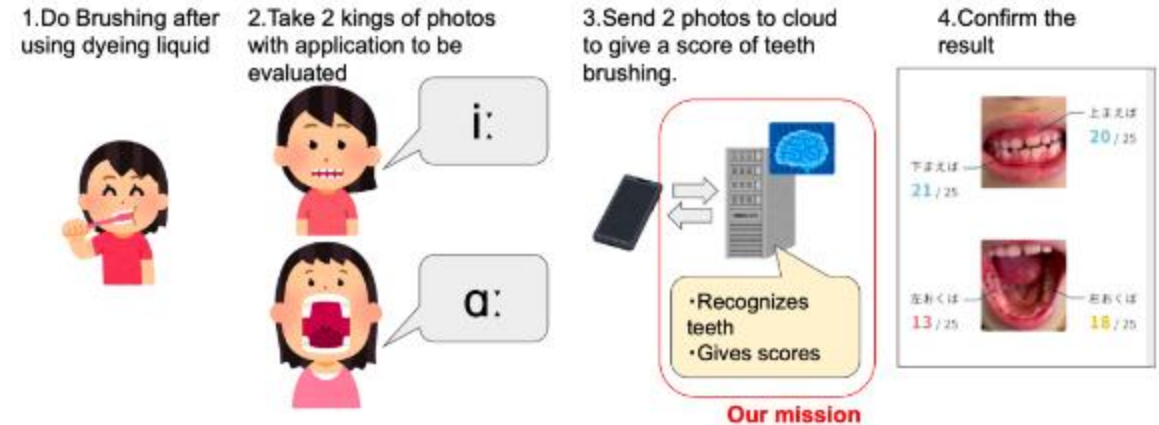
The solution was built for Dental center in Japan to analyze and score the teeth of the patients who come to the dentist, especially checking the level of plaque on their teeth by taking photos from smartphone

Values and Achievements:

- Tooth recognition accuracy: >95%
- Tooth scoring accuracy: >93%, compare with results from dental professionals
- Patients can take photo and scoring themselves at home with the app on their smartphone

Technologies:

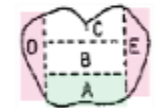
- Machine Learning & AI
- Image Processing with OpenCV & Scikit-image
- AWS Cloud service (EC2, Lambda)



Masks and indexes of each tooth

<PHP-M (Modified Patient Hygiene Performance) >

- 歯の頬側面を5分割
- A : 歯頸部 (歯肉側1/3)
- D : 歯間部 (遠心側)
- E : 歯間部 (近心側)



(出典)
Marters & Meskin: An
Innovative Technique
for Assessing Oral
Hygiene. J Dent. 1972

5. Lung Cancer Detection



Highlights suspicious regions directly on chest X-rays

AI for Lung Cancer Detection via Chest X-rays- Lunit INSIGHT CXR (Korea)

Context

- Lung cancer: Leading cause of cancer deaths in South Korea (~35,000 cases/year).
- High smoking rates drive need for effective screening.

Key Features & Benefits:

- **Malignancy Risk Scoring:** Assigns risk scores to highlight potentially malignant cases for faster prioritization.
- **Visual Assistance:** Highlights suspicious regions directly on chest X-rays to aid radiologist decision-making.
- **PACS Integration:** Easily integrates with hospital systems, ensuring smooth workflow with minimal disruption.
- **High Accuracy:** Achieves ~94% sensitivity in detecting malignant nodules (Radiology, 2020).
- **Faster Turnaround:** Reduces radiologist review time by up to 30%, enabling quicker diagnosis.
- **Early Detection Support:** Helps identify cancer at earlier stages (Stage 1/2), improving patient survival rates.
- **Global Trust:** CE marked, FDA-cleared, and deployed in over 40 countries worldwide.

Technologies: Lunit INSIGHT CXR, a deep learning-based AI system

- Detects abnormalities like nodules, masses, and consolidations.
- Assigns malignancy risk scores to prioritize urgent cases.

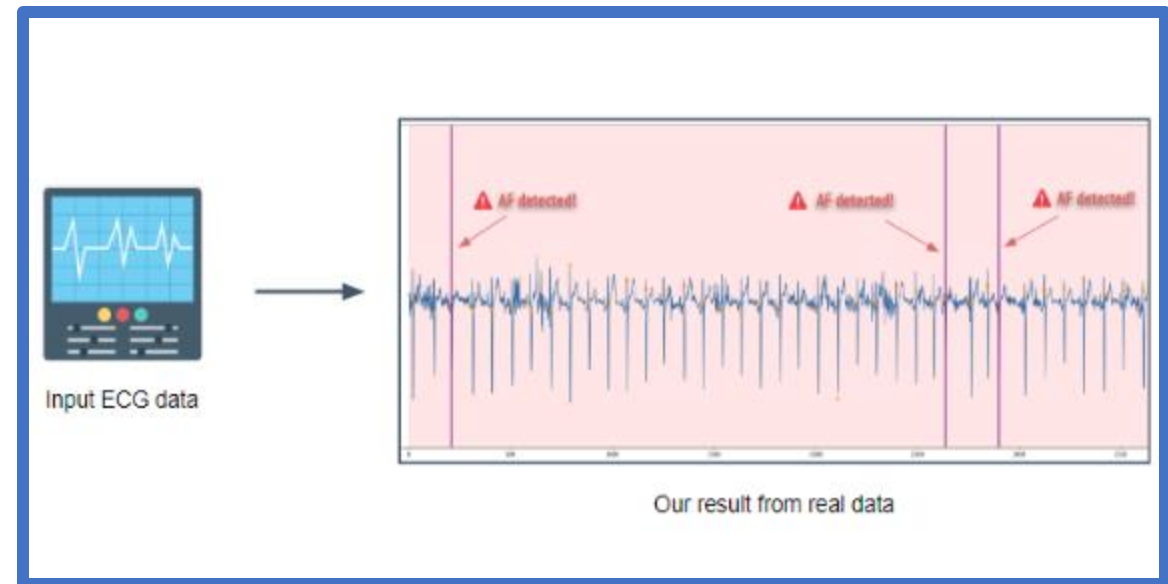
6. Stroke Assessment and Detection - TMA

Enhances stroke prevention using a comprehensive assessment tool, AI-driven predictive analytics, and continuous monitoring to detect atrial fibrillation (AFib) and mini-strokes early.

Data: Patient assessment scores, 24/7 vital sign monitoring (heart rate, blood pressure, oxygen), medical history, and AFib episode data.

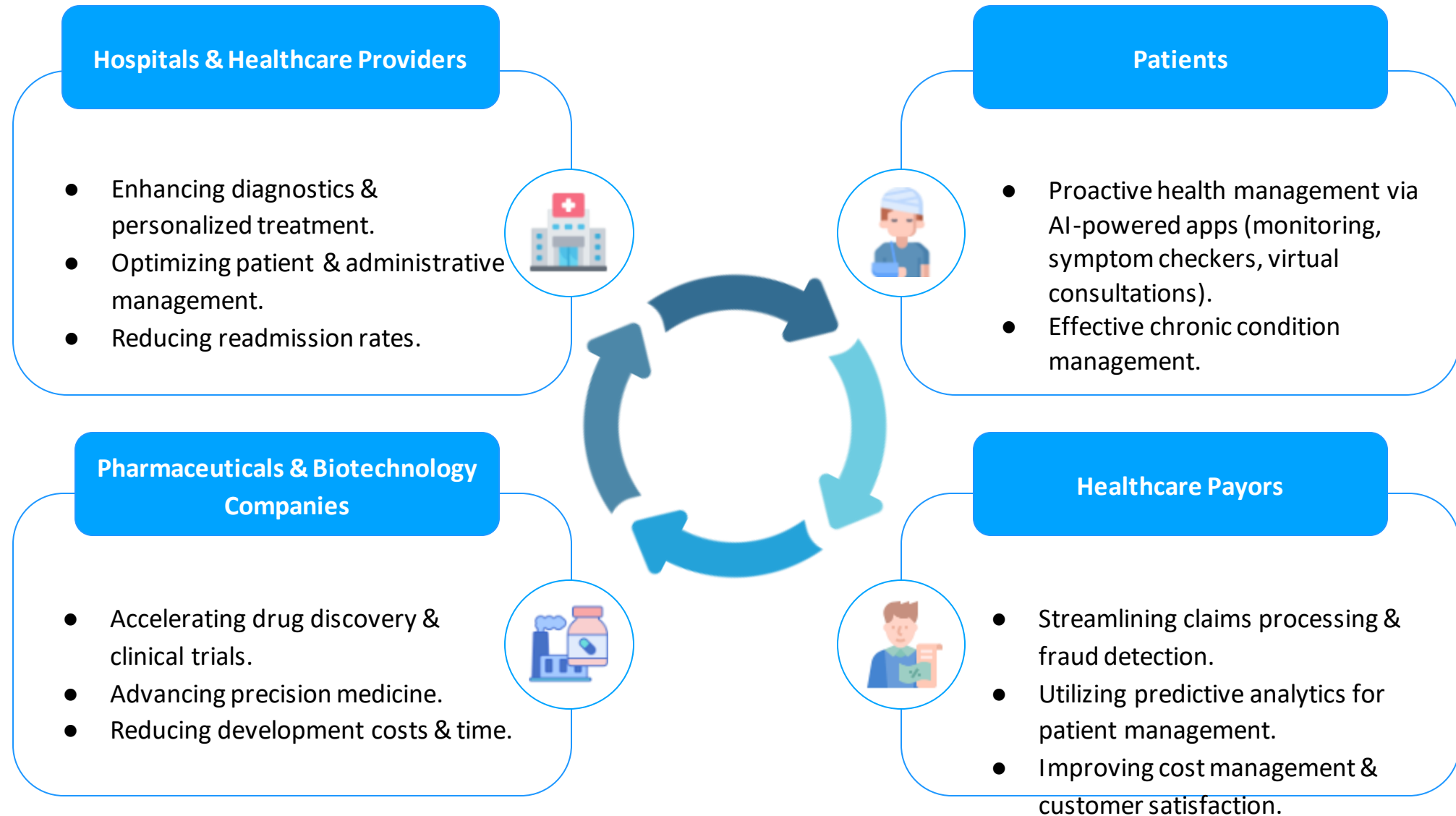
Method: Integrates real-time stroke assessment tools, AI trained on historical vital sign data for stroke risk prediction, and continuous monitoring for AFib and mini-stroke detection.

Applications: Identifies stroke risk factors, detects AFib to prevent strokes, monitors mini-stroke symptoms, and provides predictive analytics for healthcare providers.



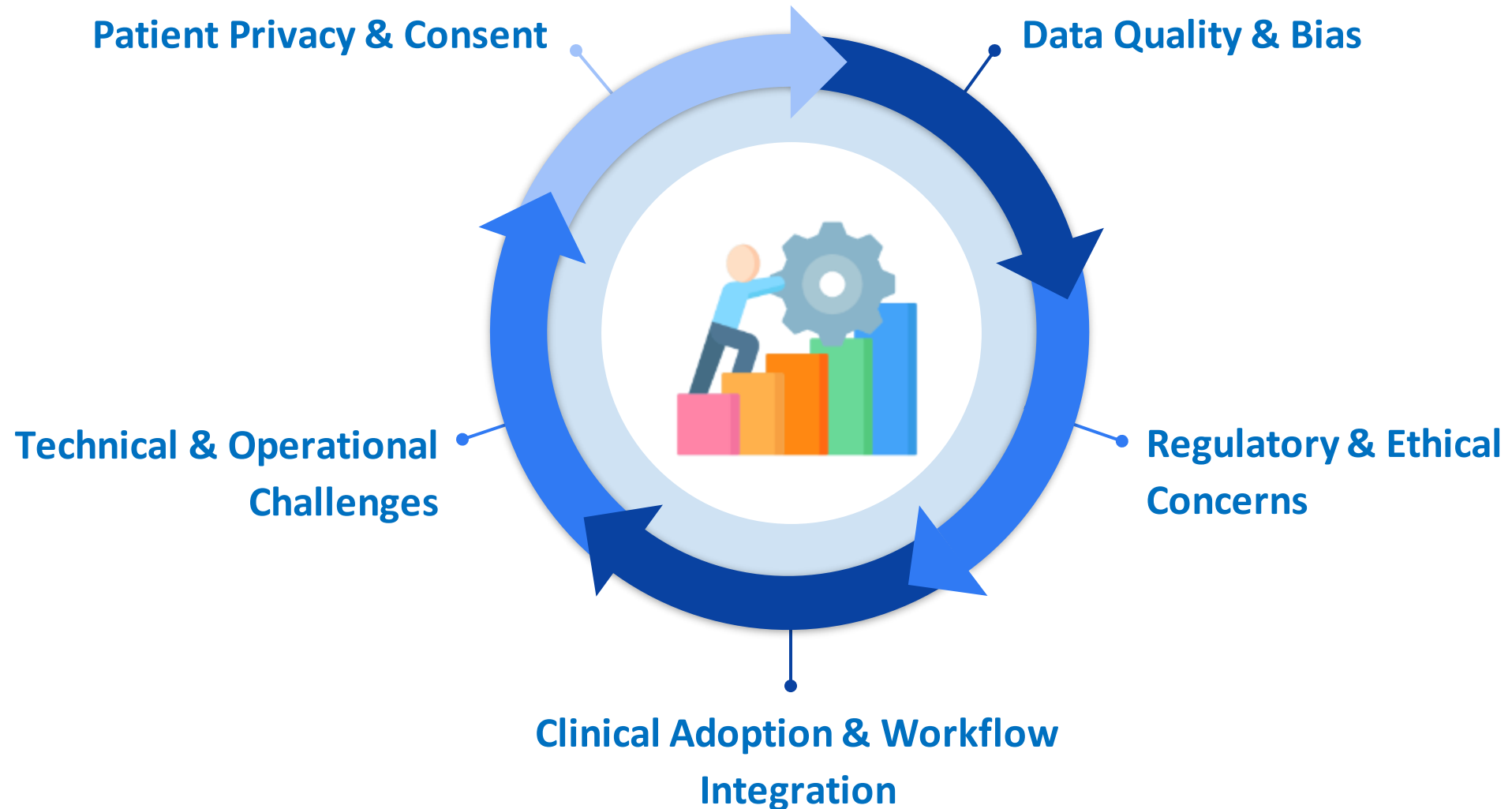
Benefits of AI in Healthcare

Who Benefits from AI in Healthcare?



Challenges and Limitations

Challenges and Limitations



Responsible AI Considerations

Ethics & Fairness

- Consistent performance across populations
- Transparent decisions
- Human oversight

Privacy & Security

- HIPAA, GDPR, HITRUST compliance
- RBAC, Data governance (PHI, PII), and data lineage
- Secure data handling
- De-identification techniques

Clinical Validation

- Real-world testing
- Ongoing monitoring
- Standards evaluation
- Transparency and EXplainability (XAI)

Standard & Compliance

Ensuring the Highest Level of Trust in **Healthcare Technology**

HIPAA

Protecting sensitive patient information in compliance with **U.S. regulations**

GDPR

Ensuring full compliance with **European data protection laws**, safeguarding patient privacy across borders.

HL7

Ensuring **interoperability** across diverse healthcare systems.

FHIR

Facilitating smooth **health data exchange** between systems for better care

PACS & DICOM

Securely managing and **sharing medical images**.



HL7® FHIR



The Trustworthy Responsible AI Network (TRAIN)

Consortium Goals

Safe, fair AI Accessible to all healthcare organizations.

Frameworks & Best Practices

Responsible AI implementation and governance.

Ethics & Bias

Address ethical concerns and reduce AI bias.

Founding Members

Mayo Clinic, Cleveland Clinic, Intermountain Health, Mass General Brigham, Microsoft.

reference: [link](#)



Future of AI in Healthcare

Healthcare AI Trends

	Generative AI in Healthcare	• Elevates diagnostics to unprecedented precision
	Agentic AI	• Exploring AI agents that can analyze patient data, medical images, and test results to speed up diagnoses and support clinicians more autonomously.
	Personalized Medicine	• Tailors treatments for individual triumphs. • AI's crucial role in creating personalized plans.
	Virtual and Augmented Reality in Healthcare	• Smart devices incorporating AI for diagnostics, monitoring, and therapeutic purposes.
	Digital Twins in Healthcare	• Predicts treatment outcomes, personalized care, and trains surgeons.



Conclusion



Conclusion

- AI in Healthcare is Real and Impactful: Artificial Intelligence is rapidly transforming healthcare.
- Achieving business value with AI requires strategic implementation, addressing challenges like technical complexities (e.g., legacy system integration), data quality and bias, and organizational readiness. Not all AI is suitable without careful consideration of these factors.
- Essential skills for AI Healthcare solution development:
 - Mastering data protocols like HL7 FHIR and DICOM is fundamental for interoperability and leveraging critical data types.
 - Proficiency in AI techniques (Machine Learning, Deep Learning, NLP, Generative AI) and Foundation Models is essential for developing impactful solutions. This also involves skills like image embedding, adapting, and fine-tuning models for specific tasks, and connecting modalities.
 - Adhering to Responsible AI Considerations emphasizing ethics, fairness, privacy, security, and clinical validation is crucial for building trustworthy systems.



Q&A

